

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

5. Project Development Phase

The Project Development Phase is where the actual FitBuddy application is implemented. The development was carried out using Python and the technologies selected during the planning phase.

First, the FastAPI backend was created. The main application initializes the FastAPI application, connects the required routes, mounts static files, and initializes the database when the application starts.

The database was implemented using SQLite and SQLAlchemy. A User model was created to store user details and generated fitness information. The application stores the original workout plan so that it can later be used when processing feedback.

Pydantic schemas were created to validate the input received from users. The application accepts information such as User ID, name, age, weight, fitness goal, and workout intensity.

The Gemini AI integration was then implemented. The workout generator receives the user's fitness information and sends a structured prompt to Gemini. The response is converted into a structured workout plan containing seven days of exercises.

A separate AI module generates nutrition and recovery recommendations. This provides an additional output along with the workout plan.

The feedback functionality was then developed. When a user submits feedback, FitBuddy retrieves the corresponding user and original workout plan from the database. The feedback and user context are sent to the AI module. The system generates a complete revised 7-day plan and stores it as the updated plan.

The frontend was implemented using Jinja2 HTML templates and CSS. The input page allows users to enter their information, while the result page displays the generated plan and nutrition recommendations.

The application also contains API routes for generating workouts, submitting feedback, retrieving users, and other operations. A demo mode was included so the application can run with sample responses during development.

Therefore, the development phase integrates all major components into one working application:

User Input → Validation → Gemini AI → Workout/Nutrition → Database → Feedback → Updated Plan